

Stage III Topology Pass on Frozen `dataset_iter035`

Run ID: `topology_pass_dataset_iter035_v1`

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Abstract

This report documents an offline topology classification and compute-cost audit on the frozen thermodynamic active-learning dataset `dataset_iter035`. The original thermodynamic labels, exact-oracle outputs, acquisition logic, and active-learning loop were not modified. The goal is to compute Pfaffian $\mathbb{Z}_2$ diagnostics and a full-Brillouin-zone bulk quasiparticle gap on existing superconducting exact points, then identify coverage gaps and candidate seeds for a future topology-aware acquisition stage. The sparse topology scatter and Delaunay edges reported here are diagnostic seeds only; they are not final topological phase boundaries.

1 Scope and Inputs

The frozen thermodynamic dataset contains 7434 exact points:

- normal: 1867,
- uniform-SC: 715,
- FFLO: 4852.

Only the existing superconducting points were passed through the topology oracle. Normal points are retained in the derived output with `topology_label = not_applicable`. No new Δ - q exact search or active-learning iteration was run.

The main generated artifacts are:

- `tables/dataset_iter035_topology_ground_v1.csv`,
- `tables/dataset_iter035_topology_ground_v1.npz`,
- `topology_pass_summary.json`,
- `topology_benchmark_report.json`,
- `topology_validation_report.md`,
- this LaTeX report and its compiled PDF.

Parquet output was requested by the Stage III plan but was not written in this local environment because neither `pyarrow` nor `fastparquet` is installed. The CSV and NPZ outputs contain the full derived dataset.

2 BdG and Pfaffian Convention

The topology pass reuses the project BdG Hamiltonian builder for the full-Brillouin-zone bulk gap. The analytic Pfaffian convention was not chosen by inspection alone; it was cross-checked against a numerical Pfaffian built from the project BdG Hamiltonian in the current Nambu/Majorana convention.

The verified convention is

$$P_0 = \left(\mu - t \cos \frac{q}{2} \right)^2 + \Delta^2 - \alpha_y^2 \sin^2 \frac{q}{2} - \left(J_A \cos \frac{q}{2} + \alpha_z \sin \frac{q}{2} \right)^2, \quad (1)$$

$$P_\pi = \left(\mu + t \cos \frac{q}{2} \right)^2 + \Delta^2 - \alpha_y^2 \sin^2 \frac{q}{2} - \left(J_A \cos \frac{q}{2} + \alpha_z \sin \frac{q}{2} \right)^2. \quad (2)$$

For gapped and reliable superconducting points,

$$P_0 P_\pi < 0 \Rightarrow \mathbb{Z}_2 = 1, \quad P_0 P_\pi > 0 \Rightarrow \mathbb{Z}_2 = 0.$$

The dimensionless Pfaffian margin is

$$m_{\text{Pf}} = \frac{\min(|P_0|, |P_\pi|)}{E_{\text{scale}}^2},$$

where E_{scale} is constructed from the model energy scales rather than set by an arbitrary fixed unit.

The analytic and numerical Pfaffian product signs agreed on all 104 non-boundary validation points. This check is a gate: the full pass should not be interpreted if this product-sign agreement fails.

3 Bulk-Gap Oracle and Resource Decision

The bulk quasiparticle gap is computed as

$$E_g = \min_{k,n} |E_n(k)|,$$

over one full Brillouin zone. The implementation uses float64/complex128, chunked batched eigensolves, and does not store full spectra for all points. The final coarse grid used $N_k = 2048$. For small-gap, small-Pfaffian-margin, and 5% random-control points, the pass performed both N_k doubling and bounded local minimization around the coarse minimum.

The 256-point pilot benchmark selected the GPU backend by measurement, not by assumption. CPU/GPU agreement at $N_k = 2048$ had maximum absolute bulk-gap difference 7.77×10^{-16} . The full bulk-gap scan recorded in the summary took 5.342 seconds on the RTX 4090 backend; the complete script, including local k refinement and report outputs, ran in roughly 70 seconds.

4 Topology Classification Results

Table 1: Topology label counts by thermodynamic phase.

Thermodynamic phase	Topology label	Count
uniform-SC	trivial	715
uniform-SC	topological	0
uniform-SC	gapless_SC	0
uniform-SC	topology_unresolved	0
FFLO	trivial	3127
FFLO	topological	1515
FFLO	gapless_SC	195
FFLO	topology_unresolved	15

Across all 5567 superconducting points, 5357 points have trusted topology labels under the current bulk-gap and exact-label reliability rules. The 195 **gapless_SC** points are points where the local k -refined bulk-gap oracle resolved the spectrum as gapless under the current tolerance. The 15 **topology_unresolved** points remain numerical-reliability states and must not be interpreted as a physical phase.

5 Coverage and Boundary-Seed Diagnostics

Trusted gapped FFLO points were used to construct a Delaunay diagnostic graph in normalized $(k_B T, J_A)$ space. This graph is used only to identify candidate topology-boundary seeds and sampling holes.

Table 2: Topology-boundary seed and coverage diagnostics.

Diagnostic	Value
Trusted FFLO gapped points	4642
Delaunay edges	13575
Delaunay triangles	8934
$\mathbb{Z}_2$ -change candidate edges	182
P_0 sign-change candidate edges	182
P_π sign-change candidate edges	0
Large circumradius coverage-hole triangles	447
Nearest-neighbor p95 distance, normalized	0.0117625
Nearest-neighbor max distance, normalized	0.0549629

All candidate $\mathbb{Z}_2$ -change edges are associated with P_0 sign changes in this diagnostic graph. This supports a topology-aware follow-up, but it does not by itself define a final contour.

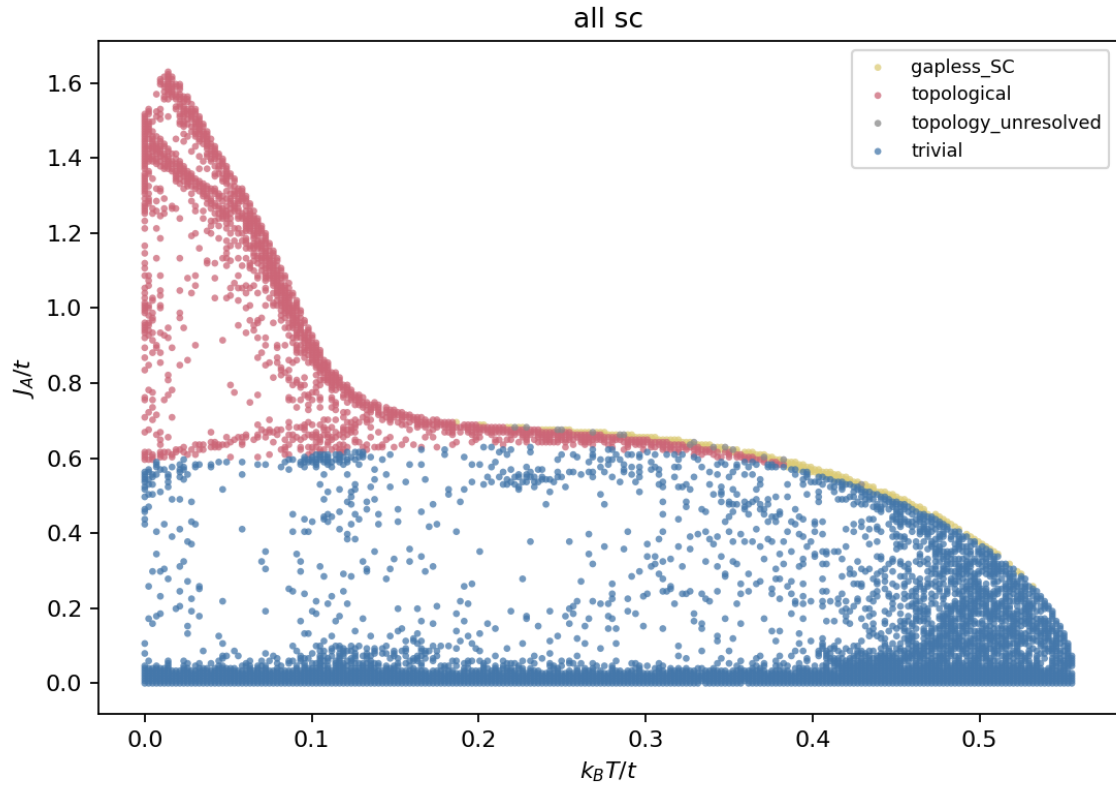


Figure 1: Offline topology labels for all superconducting points. This is a scatter diagnostic, not an interpolated final topological phase boundary.

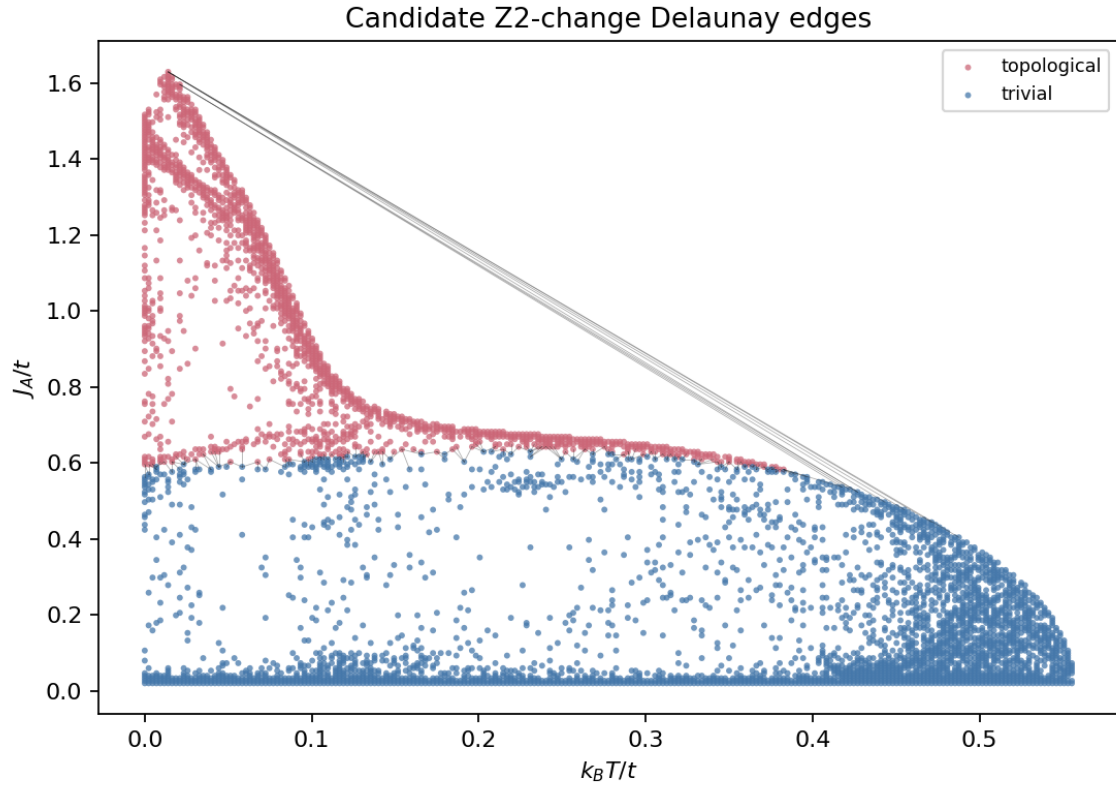


Figure 2: Candidate $\mathbb{Z}_2$ -change Delaunay edges among trusted gapped FFLO points. These edges are seeds for targeted topology refinement.

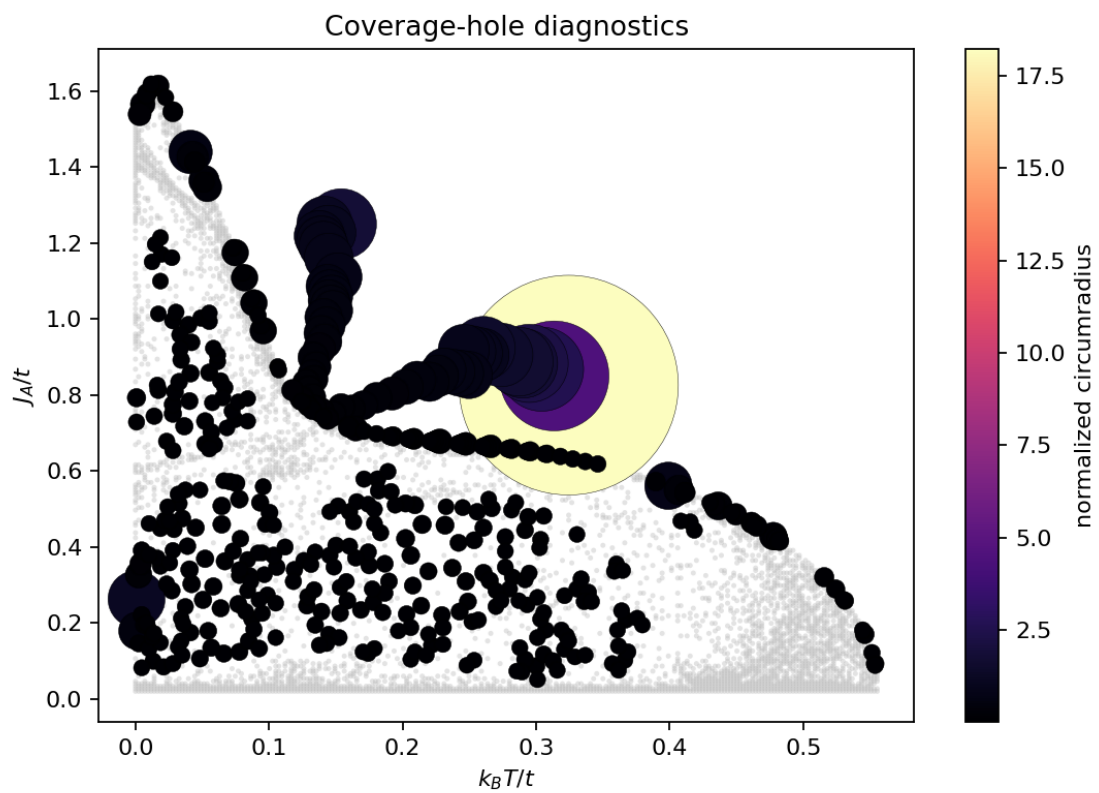


Figure 3: Large-circumradius Delaunay triangles used as sampling coverage-hole diagnostics.

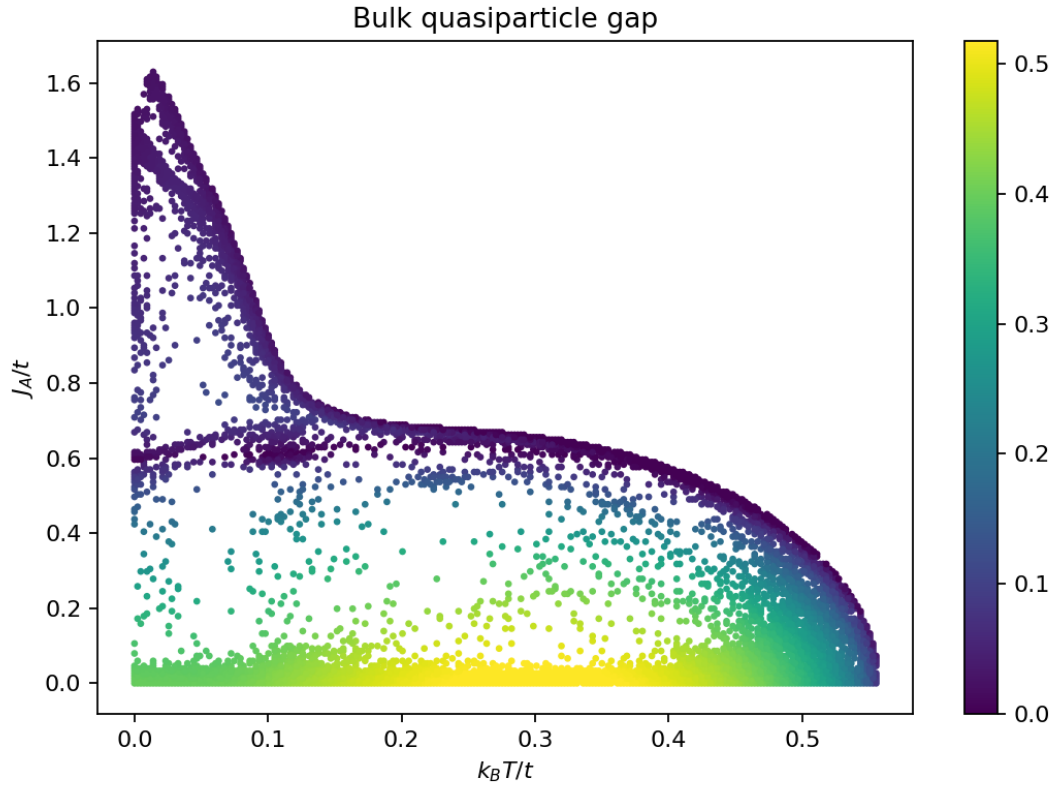


Figure 4: Full-Brillouin-zone bulk quasiparticle gap over superconducting points.

6 Caveats

- Do not modify or reinterpret the original normal / uniform-SC / FFLO labels from this topology pass.
- Do not call sparse Delaunay edges final topology contours.
- Do not treat `topology_unresolved` as a gapless phase.
- Do not use the Pfaffian sign without the full bulk-gap status.
- Hard-risk or provisional exact points are diagnostics only unless separately audited.
- The current result is single-dataset and single-pass; it is not yet a multi-seed benchmark.

7 Decision

The result falls under Case B of the Stage III plan:

Trusted topological and trivial FFLO points exist, but inherited sampling is not dense enough near candidate topology boundaries to claim a final topological contour.

The recommended next calculation is a topology-aware acquisition/refinement pilot that targets:

- small Pfaffian margin,

- small bulk gap,
- candidate $\mathbb{Z}_2$ -change Delaunay edges,
- large coverage-hole regions.

This should not restart the thermodynamic active-learning loop. It should operate as a topology-focused follow-up on the frozen thermodynamic phase map.